

Strings gnc

Characters and Strings

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Content

- Introduction to characters and strings
- String conversion functions
- String manipulation functions



Introduction

- Strings are used for storing text/characters.
- For example, "Hello World" is a string of characters.
- Unlike many other programming languages, C does not have a String type to easily create string variables.
- Instead, you must use the char type and create an array of characters to make a string in C:

```
char greetings[] = "Hello World!";
```

Another Way Of Creating Strings

- In the previous examples, we used a "string literal" to create a string variable.
- This is the easiest way to create a string in C.
- You should also note that you can create a string with a set of characters. This example will produce the same result as the previous example .

```
char greetings[] = {'H', 'e', 'l', 'l', 'o', ' ', 'W', 'o', 'r', 'l', 'd', '!', '\0'};  
printf("%s", greetings);
```

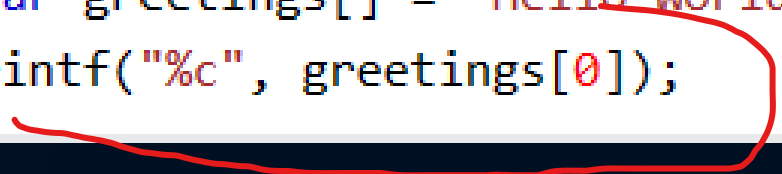
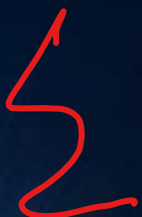
- The `\0` character at the end is known as the "null terminating character", and must be included when creating strings using this method.

Access Strings

- Since strings are actually arrays in C, you can access a string by referring to its index number inside **square brackets []**.
- This example prints the first character (0) in greetings:



```
char greetings[] = "Hello World!";  
printf("%c", greetings[0]);
```



starts at 0

Loop Through a String

- You can also loop through the characters of a string, using a `for` loop statement

```
char carName[] = "Volvo";  
int i;  
  
for (i = 0; i < 5; ++i) {  
    printf("%c\n", carName[i]);  
}
```

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sizeof() function

- Used to compute the **size of its operand**.

```
// C Program To demonstrate
// sizeof operator
#include <stdio.h>
int main(){
    printf("Size of char: %lu bytes\n", sizeof(char));
    printf("Size of int: %lu bytes\n", sizeof(int));
    printf("Size of float: %lu bytes\n", sizeof(float));
    printf("Size of double: %lu bytes\n", sizeof(double));
    printf("Size of pointer: %lu bytes\n", sizeof(void *));
    return 0;
}
```

Size of char: 1 bytes
Size of int: 4 bytes
Size of float: 4 bytes
Size of double: 8 bytes
Size of pointer: 8 bytes

```
#include <stdio.h>

int main() {
    char greetings[] = {'H', 'e', 'l', 'l', 'o', ' ', 'W', 'o', 'r', 'l', 'd', '!', '\0'};
    char greetings2[] = "Hello World!";

    printf("%lu\n", sizeof(greetings));
    printf("%lu\n", sizeof(greetings2));

    return 0;
}
```

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13

char 1
int 4
double 8
pointer 8

Cont..

string (n)

```
#include <stdio.h>
int main()
{
    int arr1[] = {1, 2, 3, 7, 98, 12, 35, 99}; // each value = 4 bytes
    printf("sizeof arr1 is :%lu ", sizeof(arr1));

    // each character = 1 byte
    char arr2[] = {'1', '2', '3', '4', '5', '6', '7', '8', '9', '\0'};
    printf("\nsizeof arr2 is :%lu ", sizeof(arr2));

    char arr3[] = "1234798012359914"; // each char = 1 byte
    printf("\nsizeof arr3 is :%lu ", sizeof(arr3));

    double arr4[] = {12.124, 12.3, 15.4}; // each value = 8 bytes
    printf("\nsizeof arr4 is :%lu ", sizeof(arr4));

    printf("\nsizeof arr4[1] is :%lu ", sizeof(arr4[1]));

    return 0;
}
```

Remember

- Size of char: 1 bytes
- Size of int: 4 bytes
- Size of float: 4 bytes
- Size of double: 8 bytes
- Size of pointer: 8 bytes

Output

```
sizeof arr1 is :32
sizeof arr2 is :10
sizeof arr3 is :17
sizeof arr4 is :24
sizeof arr4[1] is :8
```


Example of string

```
char message[] = "Good to see you,";  
char fname[] = "John";  
  
printf("%s %s!", message, fname);
```

Good to see you, John!

John

Escape character

- use the backslash escape character. To turns special characters into string characters.

Escape character	Result	Description
\'	'	Single quote
\"	"	Double quote
\\	\	Backslash

- How to print "Al-Bukhary International Universiti"

Example

It's alright.

```
char txt[] = "It\'s alright.";
```

The character \ is called backslash.

```
char txt[] = "The character \\ is called backslash.";
```

```
char txt[] = "First month is \"January\"";  
printf("%s", txt);
```

First month is "January"

Others Escape character

Escape Character	Result
\n	New Line
\t	Tab
\0	Null

③ → new line
If → Tab
10 → Null

Statement	Output
<pre>char txt[] = "Hello\nWorld!"; printf("%s", txt);</pre>	Hello World!
<pre>char txt[] = "Hello\tWorld!"; printf("%s", txt);</pre>	Hello World!
<pre>char txt[] = {'H', 'e', 'l', 'l', 'o', '\0'}; printf("%s", txt);</pre>	Hello

String Functions

- 1) String Length
- 2) Concatenate Strings
- 3) Copy Strings
- 4) Compare Strings

- 5) Others – visit [C string \(string.h\) Library Reference \(w3schools.com\)](https://www.w3schools.com/c/c_ref_string.php)
https://www.w3schools.com/c/c_ref_string.php

len
cat
cpy
cmp

strlen
strcat
strcpy
strcmp

cat
cpy
cmp

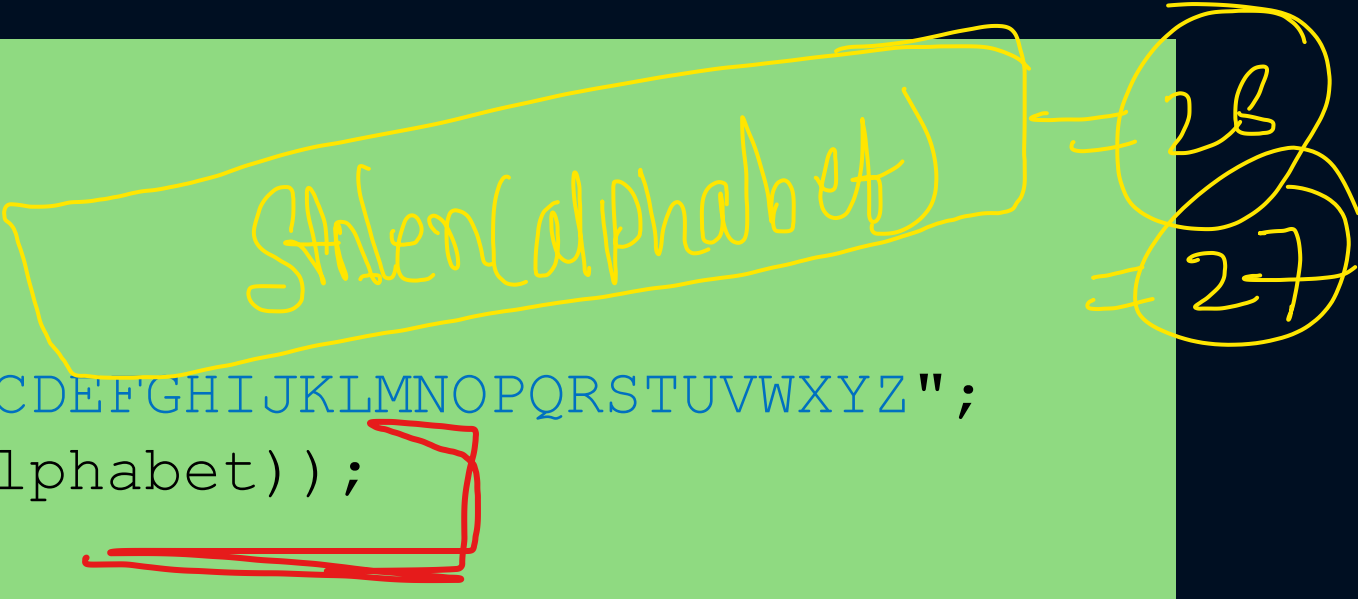
len
cat
cpy
cmp

String Length

- to get the length of a string, you can use the `strlen()` function:

```
#include <stdio.h>
#include <string.h>

int main() {
    char alphabet[] = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
    printf("%d", strlen(alphabet));
    return 0;
}
```



OUTPUT
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Concatenate Strings

- To concatenate (combine) two strings, you can use the `strcat()` function:

```
#include <stdio.h>
#include <string.h>

int main() {
    char str1[20] = "Hello ";
    char str2[] = "World!";

    // Concatenate str2 to str1 (the result is stored in str1)
    strcat(str1, str2);

    // Print str1
    printf("%s", str1);

    return 0;
}
```

Hello World!

Copy Strings

- To copy the value of one string to another, you can use the `strcpy()` function:

```
#include <stdio.h>
#include <string.h>

int main() {
    char str1[20] = "Hello World!";
    char str2[20];

    // Copy str1 to str2
    strcpy(str2, str1);

    // Print str2
    printf("%s", str2);

    return 0;
}
```

strcpy (copy, what)
str2

Hello World!

Note that the size of str2 should be large enough to store the copied string (20 in our example).

Compare Strings

- To compare two strings, you can use the strcmp() function.
- It **returns 0** if the two strings are **equal**, **otherwise** a value that is **not 0**:

```
#include <stdio.h>
#include <string.h>
```

```
int main() {
    char str1[] = "Hello";
    char str2[] = "Hello";
    char str3[] = "Hi";
```

```
    // Compare str1 and str2, and print the result
    printf("%d\n", strcmp(str1, str2));
```

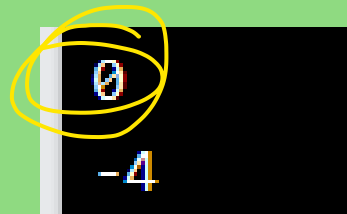
```
    // Compare str1 and str3, and print the result
    printf("%d\n", strcmp(str1, str3));
```

```
    return 0;
```

```
}
```

strcmp

strcmp()



0
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len
cat
cpy
cmp

Thank You

len
cat
cpy
cmp

Ans